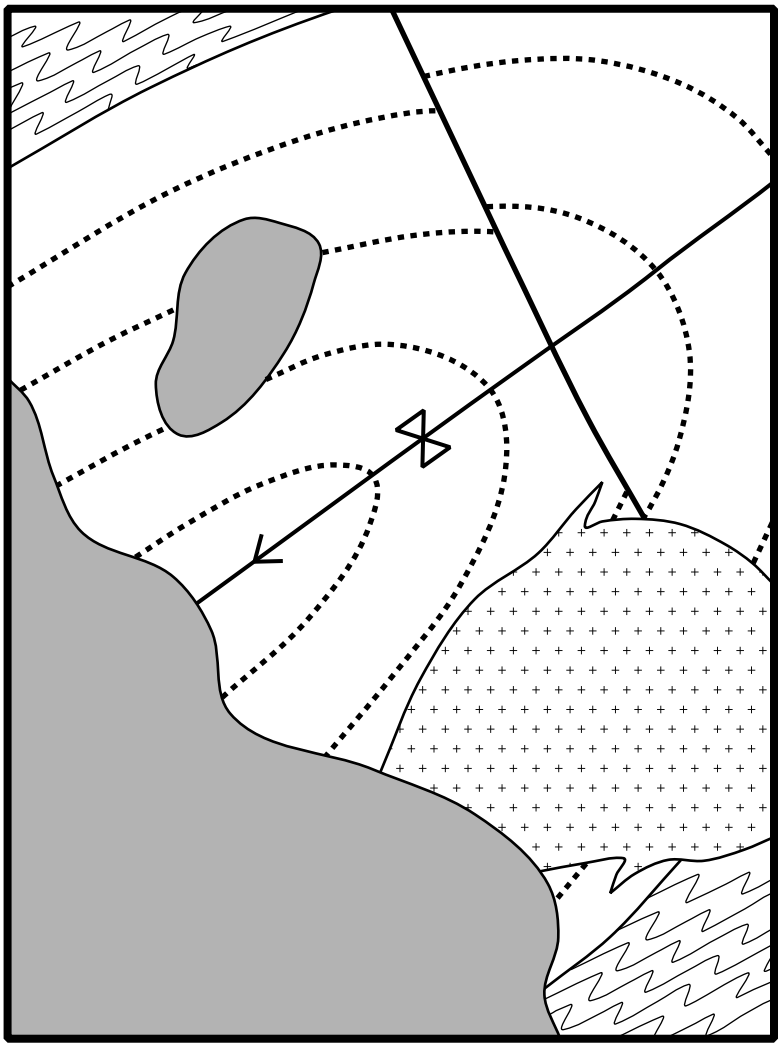
WRITING A SEQUENCE OF GEOLOGICAL EVENTS

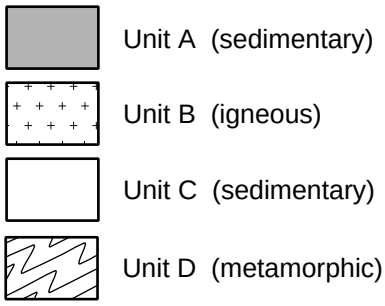
• A sequence of geological events should be a concisely worded set of bullet points, numbered to indicate the order of events

• Generally, the evidence used to deduce the sequence of events should be made explicit in the relevant bullet-points, and any quantitative information about the geometry of the structures and surfaces present should also be stated



Below is a sequence of events for the map shown left as deduced from cross-cutting relationships.

Note that without information about the topography and/or the orientation of the surfaces shown on the map, the geometry of the structures cannot be rigorously described.



Sequence of events (oldest to youngest)

1. Deposition of sedimentary precursors of Unit D
2. Folding and metamorphism of Unit D (only unit metamorphosed; early phase of folding implied by ornamentation which shows minor folds of the same asymmetry on each limb of the later synform)
3. Uplift and erosion (truncated ornamentation at the Unit C and D contact implies it is an unconformity)
4. Deposition of Unit C (Unit C has not been metamorphosed but it has been folded)
5. Folding (Unit C has been folded but the fault has not; fold is a synform plunging SW)
6. Faulting (fault offsets fold limbs in Unit C but has not affected Unit B; fault trends NW-SE, the apparent offsets suggest that it is a pure dip-slip fault with the NE side being the downthrown side)
7. Intrusion of Unit B (Unit B is not faulted; it is a plutonic intrusion)
8. Erosion (Unit A truncates the contact between Units B and C)
9. Deposition of Unit A (Unit A unconformably overlies all other units)
10. Erosion to present topography (dip of Unit A unknown − possibility of late tilting cannot be excluded)